

Integration Cheat Sheet — Worked Problems (Indefinite, Riemann, Definite, Improper)

This sheet contains **rules + step-by-step worked problems** in a clean exam-style format.

1. Indefinite Integrals

Example 1

Problem:

$$\int (4x^3 - 2 \sin x + e^{2x}) dx$$

Step 1. Integrate each term separately: 1. $\int 4x^3 dx = \frac{4x^4}{4} = x^4$ 2. $\int -2 \sin x dx = -2(-\cos x) = 2 \cos x$ 3. $\int e^{2x} dx = \frac{1}{2}e^{2x}$

Step 2. Combine results:

$$\int (4x^3 - 2 \sin x + e^{2x}) dx = x^4 + 2 \cos x + \frac{1}{2}e^{2x} + C$$

Example 2: Substitution

Problem:

$$\int 2x \cos(x^2) dx$$

Step 1. Substitution: Let $u = x^2 \Rightarrow du = 2x dx$.

Step 2. Replace in integral:

$$\int 2x \cos(x^2) dx = \int \cos(u) du$$

Step 3. Integrate:

$$\int \cos(u) du = \sin(u)$$

Step 4. Back-substitute:

$$\sin(x^2) + C$$

Example 3: Integration by Parts

Problem:

$$\int x e^x dx$$

Step 1. Formula: $\int u dv = uv - \int v du$.

Take $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$.

Step 2. Apply formula:

$$\int x e^x dx = x e^x - \int e^x dx$$

Step 3. Simplify:

$$= x e^x - e^x + C$$

2. Riemann Sums

Example: Deriving Area Under Line

Problem: Compute area under $f(x) = x$ on $[0, 1]$ using Riemann sums.

Step 1. Partition: Divide interval into n equal parts, each of width $\Delta x = 1/n$.

Step 2. Sample points: Right endpoints: $x_i = i/n$.

Step 3. Sum:

$$S_n = \sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \frac{i}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{i=1}^n i$$

Step 4. Use formula: $\sum_{i=1}^n i = n(n+1)/2$.

$$S_n = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}$$

Step 5. Limit:

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{2}$$

So $\int_0^1 x dx = 1/2$.

3. Definite Integrals

Example 1: Basic Property

Problem:

$$\int_0^{\pi} \sin x \, dx$$

Step 1. Find antiderivative: $\int \sin x \, dx = -\cos x$.

Step 2. Apply limits:

$$[-\cos x]_0^{\pi} = (-\cos \pi) - (-\cos 0)$$

Step 3. Simplify:

$$= -(-1) - (-1) = 1 + 1 = 2$$

Answer: 2.

Example 2: Area Between Curves

Problem: Find area between $y = x^2$ and $y = x$ on $[0,1]$.

Step 1. Formula:

$$A = \int_0^1 (x - x^2) dx$$

Step 2. Compute:

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

Step 3. Simplify:

$$= \left(\frac{1}{2} - \frac{1}{3}\right) - 0 = \frac{1}{6}$$

Answer: $A = 1/6$.

4. Improper Integrals

Example 1: Convergence Test

Problem:

$$\int_1^{\infty} \frac{1}{x^p} dx$$

Step 1. Case: If $p > 1$, it converges. If $p \leq 1$, it diverges.

Step 2. Proof:

$$\int_1^{\infty} \frac{1}{x^p} dx = \lim_{b \rightarrow \infty} \left[\frac{x^{1-p}}{1-p} \right]_1^b$$

If $p > 1$, exponent negative, limit goes to 0:

$$= \frac{1}{p-1}$$

If $p \leq 1$, denominator zero or limit infinite $\rightarrow$ diverges.

Example 2: Oscillatory Function

Problem:

$$\int_0^{\infty} e^{-x} dx$$

Step 1. Antiderivative: $-e^{-x}$.

Step 2. Apply limits:

$$[-e^{-x}]_0^{\infty} = (0) - (-1) = 1$$

Answer: 1.

Practice Problems (Try Yourself)

1. $\int (3x^2 + 5 \cos x) dx$
 2. $\int_0^2 (2x + 1) dx$
 3. $\int_0^\infty e^{-2x} dx$
 4. Use Riemann sum to derive $\int_0^1 x^2 dx$.
 5. Area between curves: $y = \sin x$ and $y = \cos x$ on $[0, \pi/4]$.
-

 This format keeps **every solution step-by-step, clean, and exam-ready.**